



Wind Tunnel

GROUP 8

TEST & CALIBRATION

NO.

8421.0000

VERSION / DATE / NAME

06 / 02.2020 / Zi



DESCRIPTION

The Wind Tunnel type 8421 is basically designed as test equipment for meteorological sensors.

Typical application would be test and calibration of wind speed sensors. Its open construction with the measuring section in the suction part enables compact dimensions and therefore operation in relatively small rooms.

MECHANICAL DESIGN

The mechanical design is shown on fig. 1 (page 2):
The intake section consists of the intake nozzle (1) and the rectifier part (2) with one rectifier and two screens. Another nozzle (3) – which reduces the Wind Tunnel's cross section – is placed between intake section and measuring section (4). An acrylic glass door (5), a bushing $\varnothing$ 20 mm (6) – placed in an interchangeable flange – and a conduit fitting M16 (7) serve for installation and connection of sensors to be tested. Further three conduit fittings M20 (8) can receive Prandtl tubes or thermoelectric anemometers for reference measurement. The fan (9) is direct mounted on the motor shaft. The whole equipment is carried by a number of rubber shock mounts. The fan's exhaust cross section is covered by a protection lattice (10).

DRIVE SYSTEM

The Asynchron motor (11) is SCR-controlled in order to achieve a wide velocity range operation. The 19"-control cabinet, containing the speed control (refer to fig. 2) comprises the following parts:

- (13) Indication motor current
- (14) Indication main voltage
- (15) Emergency stop
- (16) Reset emergency stop
- (17) 1 manual / 2 automatic switch
- (18, 20) ON and OFF switches (push buttons)
- (19, 21) Control lamps
- (22) Reset alarm
- (23) Operating hours
- (24) Speed control unit (manual operation)
- (25) Indication rotational speed of motor by LED-display
- (26) Main switch
- (27) Power sockets
- (28, 29) Sensor signal input tableau
- (30) Automatic operation

Devices for measurement of room temperature, air pressure and relative humidity as well as computation of the true air density and exact air velocity in the Wind Tunnel completes the system.

Technical data are subject to change!



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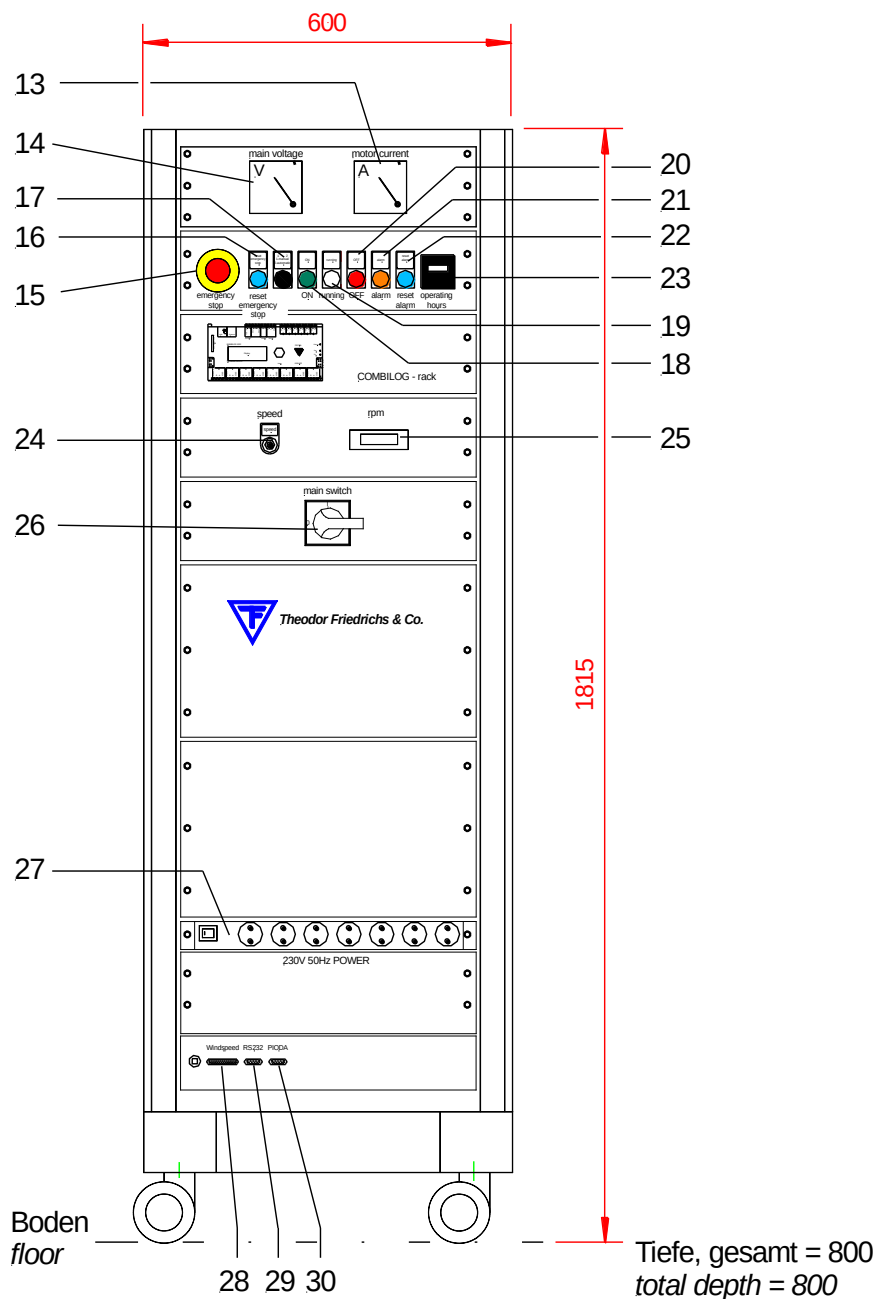
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FIG. 2: 19" CONTROL CABINET



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TECHNICAL DATA

Design:	Open-circuit Wind Tunnel
Measuring range:	0.15...approx. 50 m/s*
According Reynolds number:	$0.65 \times 10^4 \dots 2.15 \times 10^6$ (at Pa = 1013 hPa and t = 20 °C)
Degree of turbulence:	at v = 2 m/s ... 50 m/s: 1 % measured in working section (radius of 175 mm from tunnel centre)
Drive Power:	58 kW*
Drive control:	S C R
Power supply:	400 V, 50 Hz AC/3 Ph
Dimensions:	Refer to dimension sketch

WEIGHT

Wind Tunnel incl. drive:	approx. 1400 kg
Control cabinet:	approx. 275 kg

***) At ambient temperature 20 °C and 1013 hPa and nominal operating temperature at the motor.**

REQUIREMENTS FOR INSTALLATION

HANDLING

The wind tunnel will be delivered in 5 wooden boxes. The weight of the heaviest box is approx. 1.070 kg. Carrying and lifting devices have to be provided accordingly.

INSTALLATION AREA

It is recommended, that the wind tunnel will be installed separately to the control rack and the operators work desk. Minimum requirements for the room size can be found on the relevant drawing. It is recommended, that both rooms are air-conditioned. The operator room shall have a window to the wind tunnel room, in order, that the operator can watch the starting threshold of the anemometer on test. If that is not possible, an observation camera system has to be provided. The entrance door of the wind tunnel room must have a minimum opening of 150 cm.

The ground floor of the wind tunnel room should be horizontal, but there is no special requirement of the max. load per square meter to be provided. The installation is on rubber chock mounts, even the wind tunnel system itself is balanced and therefore vibration free.

The legs with the rubber chock mounts have not to be fixed at the ground.

Theodor Friedrichs will prepare a drawing of the complete installation layout, if the customer provides a drawing or sketch of their available rooms.

POWER SUPPLY

Mains supply near to the installation point of the wind tunnel control rack with 3 x 400 V, 50 Hz, 60 kVA and 3 x 160 A fuses has to be provided.

The power supply cable between mains supply fuse cabinet and the control rack, will be 5 x 50 mm².

INSTALLATION

An opening between the wind tunnel room and the operator room for the various connection cables between the tunnel and the control rack must be provided.

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OPERATING INSTRUCTIONS

ATTENTION:

According to local laws or regulations a suitable sound insulation (by ear muffler or separating wall) has to be provided.

1. Installation

The Wind Tunnel has to be installed in a room separated from the operator's desk. This room should not be entered during operation with higher speed. The Wind Tunnel will be supplied in 4 sections.

The first section to be placed correctly is the Motor / Fan frame.

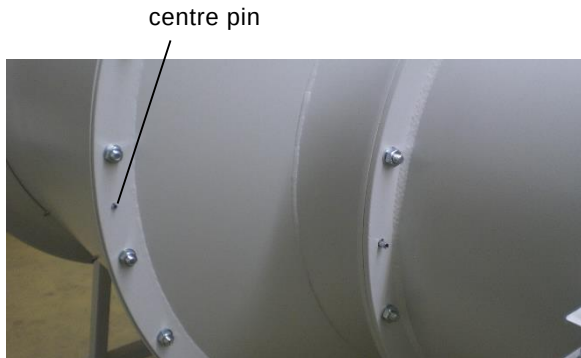
The second section is the short, straight tube, which will be fixed at the fan side of the first section.

Next section is the measuring section. After placing this section close to the second section, it must be aligned by using 2 centre pins (one on each side), before the sections can be screwed together.



centre pin

The rectifier section must be connected in the same way.



centre pin

ATTENTION:

Care has to be taken, that the wires of the three-phase mains supply will be connected in the correct sequence.

All supporting bases are mounted on vibration absorbing rubber elements. Under normal conditions screwing down to the floor will therefore not be necessary. If it is required, however, commercial size concrete wall dowels for bolts 8 mm in diameter may be used. The floor should be made of reinforced concrete.

2. Electric connections

The connection to the electric mains respectively between control cabinet and driving motor is done according to the connection diagrams, delivered with the system and must only be carried out by an expert electrician!

3. Starting operation

Before starting up it must be checked that no loose parts are left in the suction part and that the rotor can turn freely.

Operation start is done in the following sequence:

- 3.1 Set Main switch (26) to »ON« (I);
- 3.2 Press »Reset emergency stop« button; (16)
- 3.3 Blinking »Alarm« lamp shall be off; (21)
- 3.4 Select manual or automatic operation; (17)
Continue with 3.5, if manual operation selected. If automatic operation is selected, continue as described in the manual
» Sensor Calibration Program «.
- 3.5 Press button » ON «; (18)
- 3.6 » Running « indication lamp shall be lighted (19)

Turn Potentiometer (24) slowly clockwise and the rotor of the Wind Tunnel starts turning. The instantaneous value of the engine speed (RPM) is displayed on a digital meter (25). Main voltage and motor current are shown on the two analogue meters (13) and (14).



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After reaching the desired wind speed the potentiometer may be locked in position by a lever.

Turning off the Wind Tunnel is done in the inverse sequence as described above.

ATTENTION:

If during measurements the wind speed is adjusted to "zero" i.e. that the rotor comes to a standstill, it is absolutely necessary to turn off the motor by using red push button » OFF « (20) as otherwise the working life of the driving motor is reduced!

4. Maintenance

From time to time this unit should be cleaned. It must especially be taken care that the screen of the suction part is free of impurities.

ATTENTION:

Generally it has to be avoided that light, loose objects are left within the reach of the circulating air since they may be sucked in.

Maintenance work at the motor is carried out according to the supplied maintenance instructions. Prolonged standstill may reduce the lubrication properties. Corrosion might occur in bearings. In order to avoid this, the fan should be operated for about 10 minutes every month. If this is impossible at least the impeller should be turned by hand.

5. Hints for measurements

Mounting:

The Wind Tunnel has a flange plate with universal mount 20 mm in diameter. For the individual instruments under test suitable matching adapters have to be used warranting a safe operation. During operation it must always be taken care that the mounting screws are tightened well!

Maintenance cover:

During measurements the acrylic glass door must by all means be kept closed!

Arrangement of instruments:

Generally all instruments under test should be arranged so that measuring elements (cup assembly, propeller or similar) are located as close as possible to the centre of the tunnel cross section. The outer zone of the cross section of approx. 10 cm should be avoided completely because of the boundary layer and the greater turbulence occurring herein. The Prandtl tube for differential pressure measurements or thermal anemometer sondes etc. may be mounted in screw connections (M20) provided for this purpose. (Max. $\varnothing$ of sonde = 10 mm).



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Reference measurements

For measuring lowest wind speeds with high resolution a thermal anemometer (measurement range approx. 0...5 m/s) is used.

For the speed range ≥ 5 m/s differential pressure (dynamic pressure) measuring systems are used consisting of a Prandtl tube and a suitable micro-pressure gauge. For further details refer to the operating manuals of the individual instruments!

Note for determining the actual air density

In order to compute the airspeed from the measured dynamic pressure the knowledge of the true air density inside the tunnel is necessary. For this purpose the following parameters must be known:

t	=	Air temperature	(°C)
Pa	=	Air pressure	(hPa)
r.h.	=	Relative humidity	(%)

These parameters should be measured as near as possible to the Wind Tunnel!

The relevant sensors will be delivered with the system.

The true air density results from the relation:

$$\rho = \frac{PL}{T_v \cdot 2.8705}$$

$$T_v = (t + 273.16) \cdot \left[1 + \frac{E(t) \cdot r.h. \cdot 0.00622 \cdot 0.608}{Pa - 0.00378 \cdot E(t) \cdot r.h.} \right]$$

$$E(t) = 6.1078 \cdot e^{\left(\frac{17.08085 \cdot t}{2341.75 + t} \right)}$$

If the dynamic pressure q is measured in mm WC (water-column pressure) then the wind speed results from:

$$v = \sqrt{\frac{2 \cdot q \cdot g}{\rho}}$$

using:

v	=	Wind speed	(m/s)
q	=	Dynamic pressure	(mm WC $\hat{=}$ kg/m ²)
g	=	gravity	(m/s ²)
ρ	=	Air density	(kg/m ³)

REFERENCE SENSORS

TUNNEL AIRFLOW

Thermal anemometer system, part no. **8477.1000** used for airflow speed between 0.15 ... 1 m/s

Range: 0.125 ... 7.5 m/s

Accuracy: 0.5 % FS of selected range

Flow calculation system, part no. **8477.2000** used for airflow speed between 1 m/s ... 50 m/s.
4 differential pressure sensors

Range: 0 ... 25 Pa
0 ... 100 Pa
0 ... 400 Pa
0 ... 2500 Pa

Accuracy: 0.4 % FS

Air pressure

Range: 800 ... 1100 hPa

Accuracy: 0.5 % FS

AIR TEMPERATURE, PART NO. 3032.0000

Range: -40 ... +60 °C

Accuracy: 0.2 %

AIR HUMIDITY, PART NO. 3032.0000

Range: 5 ... 98 %

Accuracy: 2.5 %

PRANDTL TUBE, PART NO. 8000.0001

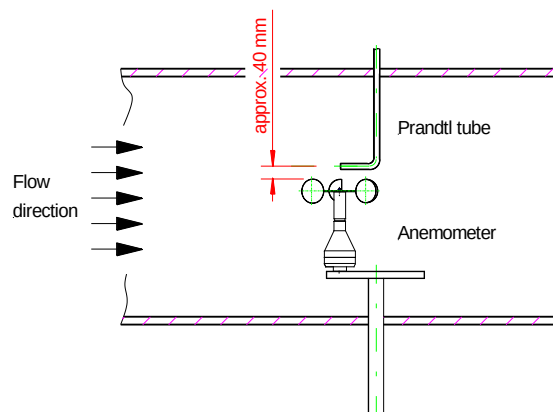
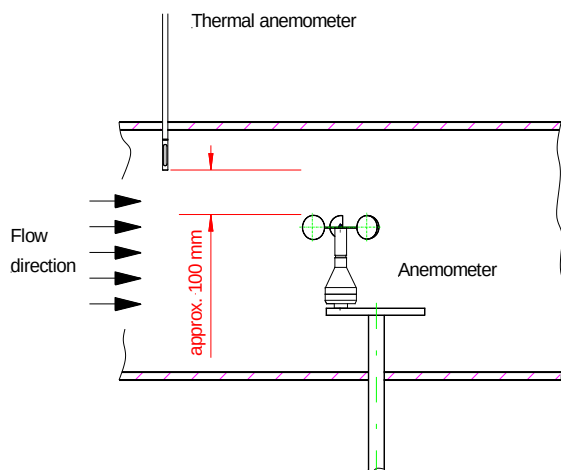
Limit of sensitivity: 0.8 m/s



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INSTRUMENTS-ON-TEST:

THERMAL ANEMOMETER AND PRANDTL TUBE POSITION



VIEW FROM TOP

